

0 6

A scientist crossed a strain of the fungus *Neurospora* producing pink spores with a strain of *Neurospora* producing white spores.

To cross these strains, he used aseptic techniques. He moved a small agar cube containing one strain of the fungus onto a new agar plate. Then he placed a second agar cube containing the other strain of fungus next to the first agar cube.

0 6**1**

Describe and explain **three** ways in which the scientist would ensure he used aseptic techniques to move each cube of agar onto a new agar plate.

[3 marks]

1

2

3

In the life cycle of *Neurospora* most stages are haploid. Fusion of two haploid strains of this fungus produces diploid zygotes. Nuclear division in these zygotes occurs by meiosis.

0 6**2**

Give **two** differences between mitosis and meiosis.

[2 marks]

1

2



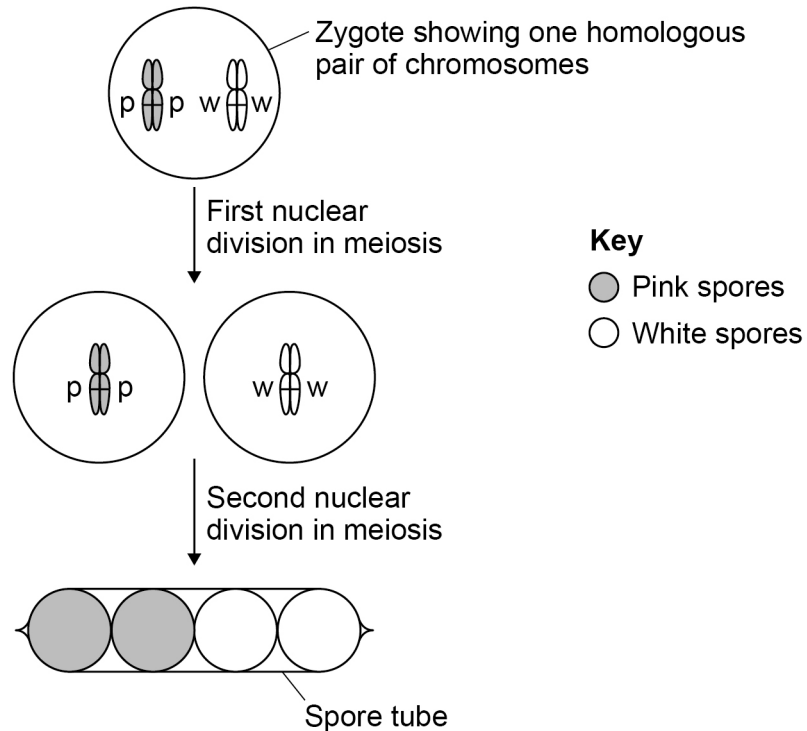
At the end of meiosis, this fungus produces cells called spores.

The spores are produced in narrow tubes that restrict their movement. As a result, each tube contains a single line of spores. The spores are coloured either pink or white.

The spore colour gene is located on a pair of homologous chromosomes. Each zygote produced in this cross has one chromosome with a pink allele (p) and one chromosome with a white allele (w).

This is shown in **Figure 7**.

Figure 7



0 6 . 3 There are seven chromosomes in a spore nucleus.

Place a tick (✓) in the box next to the number that represents the number of **chromatids** present in the zygote shown in **Figure 7**.

[1 mark]

7

☐

14

☐

21

☐

28

☐

Question 6 continues on the next page

Turn over ►



Table 2

Type of spore tube	Arrangement of coloured spores	Number of narrow tubes
1		81
2		78
3		10

Using all the information in this question, what can you conclude from the scientist's results about the movement of chromosomes in meiosis in this fungus?

[3 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.